

Magma demo 2 - September 30, 2025

Example 1: Let's find an elliptic curve in the form $y^2 = x^3 + ax^2 + bx + c$ that goes through the points

(3,6)

(4,4)

(6,6)

Each of these points (x_i, y_i) give a condition on the coefficients.

We need $a \cdot x_i^2 + b \cdot x_i + c = y_i^2 - x_i^3$

```
> x1 := 3;
```

```
> x2 := 4;
```

```
> x3 := 6;
```

```
> y1 := 6;
```

```
> y2 := 4;
```

```
> y3 := 6;
```

We want to make a matrix where the coefficients of a go in the first column, the coefficients of b in the second column, the coefficients of c in the third column, and the values of $y_i^2 - x_i^3$ in the last column.

```
> M := Matrix(Rationals(), [[x1^2, x1, 1, y1^2 - x1^3], [x2^2, x2, 1, y2^2 - x2^3], [x3^2, x3, 1, y3^2 - x3^3]]);
```

```
> M;
```

```
[ 9 3 1 9]
```

```
[ 16 4 1 -48]
```

```
[ 36 6 1 -180]
```

We ask Magma to row reduce M :

```
> EchelonForm(M);
```

```
[ 1 0 0 -3]
```

```
[ 0 1 0 -36]
```

```
[ 0 0 1 144]
```

There is a unique solution where $a = -3$, $b = -36$ and $c = 144$. This means we get the elliptic curve $y^2 = x^3 - 3x^2 - 36x + 144$.

```
> E := EllipticCurve([0, -3, 0, -36, 144]);
```

```
> E;
```

Elliptic Curve defined by $y^2 = x^3 - 3x^2 - 36x + 144$ over Rational Field

We check that those three points are on the curve.

```
> E![3,6];
```

```
(3 : 6 : 1)
```

```
> E![4,4];
```

```
(4 : 4 : 1)
```

```
> E![6,6];
```

```
(6 : 6 : 1)
```

What is the rank of E ?

```
> G, phi := MordellWeilGroup(E);
```

Here G is an abelian group and ϕ is the isomorphism from G to $E(Q)$.

```
> G;
```

Abelian Group isomorphic to $\mathbb{Z} + \mathbb{Z}$

Defined on 2 generators (free)

The group is isomorphic to $\mathbb{Z} \times \mathbb{Z}$.

* Note: Sometimes the computations to compute the rank of E are quite difficult even for Magma. The command

```
> SetClassGroupBounds("GRH");
```

tells Magma to assume the generalized Riemann hypothesis. This can make computing the rank of an elliptic curve much faster (although the answer is conditional on this hypothesis).

```
> phi(G.1);
(6 : 6 : 1)
> phi(G.2);
(3 : -6 : 1)
> P := phi(G.1);
> Q := phi(G.2);
```

We call $P = (6,6)$ and $Q = (3,-6)$ the generators.

```
> P+Q;
(10 : -22 : 1)
> P-Q;
(-6 : -6 : 1)
```

The point $(4,4)$ is on the curve. There must be a way to get $(4,4)$ from P and Q using the group law.

We will ask Magma if the points P , Q , and $(4,4)$ on E are linearly independent.

```
> IsLinearlyIndependent([P,Q,E![4,4]]);
false (-2 1 -1)
```

They are not, and we have to have $(-2)*P + 1*Q + (-1)*(4,4) = 0$. This means that $-2*P + Q$ should be $(4,4)$.

```
> -2*P+Q;
(4 : 4 : 1)
```

Example 2: Let's make an elliptic curve with positive rank that has a point of order 7 on it. How do we find an elliptic curve with a point of order 7?

It turns out that if P is a point on an elliptic curve with order ≥ 4 , then by translating and rotating the equations you can move P to the point $(0,0)$ and put E into the form

$$y^2 + axy + by = x^3 + bx^2$$

for some constants a and b .

```
> F<a,b> := FunctionField(Rationals(),2);
> E := EllipticCurve([a,b,b,0,0]);
> E;
Elliptic Curve defined by  $y^2 + a*x*y + b*y = x^3 + b*x^2$  over Multivariate rational function field of rank 2 over Rational Field
> P := E![0,0];
```

What conditions on P will mean it has order 7? We need $x(4*P)$ is equal to $x(-3*P)$.

```
> 4*P;
((-a*b + b^2 + b)/(a^2 - 2*a + 1) : (-a^2*b^2 + 3*a*b^2 - b^3 - 2*b^2)/(a^3 - 3*a^2 + 3*a - 1) : 1)
> -3*P;
(-a + 1 : a^2 - 2*a + 1 : 1)
```

So we want $(-a*b + b^2 + b)/(a-1)^2 = -(a-1)$.

This means $-a*b + b^2 + b + (a-1)^3 = 0$.

If we homogenize this we get $-a*b*c + b^2*c + b*c^2 + (a-c)^3 = 0$.

```
> R<a,b,c> := PolynomialRing(Rationals(),3);
> f := -a*b*c + b^2*c + b*c^2 + (a-c)^3;
> C := Curve(ProjectiveSpace(Rationals(),2),f);
> Genus(C);
0
```

This is a genus zero curve and we will search for points on this curve.

```
> pts := PointSearch(C,100);
```

```
> #pts;
28
```

OK we found 28 points. Let's run through all these points, and compute the ranks of the elliptic curves.

```
> for i in [1..#pts] do
for> chk, E := IsEllipticCurve([pts[i][1],pts[i][2],pts[i][2],0,0]);
```

The command above tests to see if the curve with the parameters we have given is singular or not. If it is not, then chk is true.

```
for> if chk then
for|if> printf "For i = %o, the rank of E is %o.\n",i,Rank(E);
for|if> end if;
for> end for;
For i = 1, the rank of E is 0.
For i = 2, the rank of E is 0.
For i = 3, the rank of E is 0.
For i = 4, the rank of E is 0.
For i = 5, the rank of E is 0.
For i = 6, the rank of E is 0.
For i = 7, the rank of E is 0.
For i = 8, the rank of E is 0.
For i = 9, the rank of E is 0.
For i = 10, the rank of E is 0.
For i = 11, the rank of E is 0.
For i = 12, the rank of E is 0.
For i = 13, the rank of E is 0.
For i = 14, the rank of E is 0.
For i = 15, the rank of E is 0.
For i = 16, the rank of E is 0.
For i = 17, the rank of E is 0.
For i = 18, the rank of E is 1.
For i = 19, the rank of E is 1.
For i = 21, the rank of E is 0.
For i = 22, the rank of E is 0.
For i = 23, the rank of E is 0.
For i = 24, the rank of E is 0.
For i = 25, the rank of E is 0.
For i = 26, the rank of E is 0.
For i = 27, the rank of E is 0.
For i = 28, the rank of E is 0.
```

Notice that there is no i=20 line. For that point on C, the corresponding $y^2 + axy + by = x^3 + bx^2$ is singular.

```
> pts[18];
(0 : 1 : 0)
```

That's not great. This corresponds to $b = \text{infinity}$.

```
> pts[19];
(-5/16 : 63/64 : 1)
```

This works. We have $a = -5/16$ and $b = 63/64$. Let's make that elliptic curve.

```
> E := EllipticCurve([-5/16,63/64,63/64,0,0]);
> E2 := MinimalModel(E);
> E2;
Elliptic Curve defined by  $y^2 + x*y = x^3 - 32311*x + 6205097$  over Rational Field
> G, phi := MordellWeilGroup(E2);
> G;
Abelian Group isomorphic to  $Z/7 + Z$ 
Defined on 2 generators
Relations:
7*G.1 = 0
> phi(G.1);
(422 : 8021 : 1)
> phi(G.2);
```

```
(1094 : 35237 : 1)
```

So the rank of E is 1. Every point has the form $k*(422,8021) + l*(1094,35237)$ for some k in $\{0,1,2,3,4,5,6\}$ and l an integer.

Example 3: The group of points on $y^2 = x^3 - 16x + 16 \pmod{3}$ and $\pmod{5}$.

```
> E := EllipticCurve([0,0,0,-16,16]);
```

This defines an elliptic curve over $\mathbb{Q}$. How do we get Magma to reduce it mod 3 or mod 5?

```
> E5 := ChangeRing(E,GF(5));
```

Here $GF(5)$ means "the field with 5 elements."

```
> E5;
Elliptic Curve defined by  $y^2 = x^3 + 4x + 1$  over  $GF(5)$ 
```

Magma reduced the curve mod 5. What are the points on this curve?

```
> Points(E5);
{@ (0 : 1 : 0), (0 : 1 : 1), (0 : 4 : 1), (1 : 1 : 1), (1 : 4 : 1), (3 : 0 : 1), (4 : 1 : 1),
(4 : 4 : 1) @}

```

That's the list of all the points mod 5. What is the group structure?

```
> G, mp := AbelianGroup(E5);
> G;
Abelian Group isomorphic to  $\mathbb{Z}/8$ 
Defined on 1 generator
Relations:
8*G.1 = 0
```

So $E(F_5)$ is isomorphic to $\mathbb{Z}/8\mathbb{Z}$

```
> mp(G.1);
(0 : 4 : 1)
```

The point $(0,4)$ is a generator.

```
> 2*mp(G.1);
(4 : 4 : 1)
> 3*mp(G.1);
(1 : 1 : 1)
> 4*mp(G.1);
(3 : 0 : 1)
> 5*mp(G.1);
(1 : 4 : 1)
> 6*mp(G.1);
(4 : 1 : 1)
> 7*mp(G.1);
(0 : 1 : 1)
> 8*mp(G.1);
(0 : 1 : 0)
```

We see that $(0,4)$ has order 8.

Now we look at the curve $y^2 = x^3 - 16x + 16 \pmod{3}$.

```
> E3 := ChangeRing(E,GF(3));
> E3;
Elliptic Curve defined by  $y^2 = x^3 + 2x + 1$  over  $GF(3)$ 
```

```
> Points(E3);
{@ (0 : 1 : 0), (0 : 1 : 1), (0 : 2 : 1), (1 : 1 : 1), (1 : 2 : 1), (2 : 1 : 1), (2 : 2 : 1)
@}
```

There are 7 points on the curve.

```
> AbelianGroup(E3);
```

Abelian Group isomorphic to $\mathbb{Z}/7$

Defined on 1 generator

Relations:

$7 \cdot 1 = 0$

The group of points is isomorphic to $\mathbb{Z}/7\mathbb{Z}$.

Any questions?

Eleventh day of class - 9/30

Fun fact: Can you find three rational numbers that add to zero and multiply to 6?

What about 3 rational numbers that add to zero and multiply to 67?

If

$$x = \frac{117272043}{6466424}, y = -\frac{1459264}{7082019}, z = -\frac{28654609}{1598184},$$

then $x + y + z = 0$ and $xyz = 67$.

Magma demo - See notes11.txt

Questions?

Last time we started talking about torsion subgroups. We talked about points of order 2 and 3 on elliptic curves. Today we'll talk about the complex points on an elliptic curve.

Section 2.2 - Real and complex points on cubic curves

Definition. A lattice L is a set of complex numbers of the form $\{a\omega_1 + b\omega_2 : a, b \in \mathbb{Z}\}$, where ω_1 and ω_2 are complex numbers that are linearly independent over $\mathbb{R}$.

Draw a picture.

The Weierstrass $\wp$ -function is defined by

$$\wp_L(z) = \frac{1}{z^2} + \sum_{\omega \in L} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right).$$

Fact: The series above converges for every $z \notin L$. When $z \in L$, $\wp_L(z)$ has a double pole at z . (That is, $\lim_{w \rightarrow z} (w - z)^2 \wp_L(w)$ exists and is nonzero.)

Weierstrass built this function to make it so that $\wp_L(z + \omega_1) = \wp_L(z)$ and $\wp_L(z + \omega_2) = \wp_L(z)$. Because these formulas are true, the same thing is true of the $\wp'_L(z)$, the derivative of $\wp_L(z)$.

Crazy fact: There are constants $g_2(L)$ and $g_3(L)$ so that

$$[\wp'_L(z)]^2 = 4\wp_L(z)^3 - g_2(L)\wp_L(z) - g_3(L).$$

In other words, for every complex number z except those with $z \in L$, the point $P = (\wp_L(z), \wp'_L(z))$ is a point on the elliptic curve $y^2 = 4x^3 - g_2x - g_3$.

(For all choices of L , $g_2(L)^3 - 27g_3(L)^2 \neq 0$. This means that $y^2 = 4x^3 - g_2x - g_3$ is non-singular.)

Things get better. There are addition identities:

$$\wp(u+v) = \frac{1}{4} \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2 - \wp(u) - \wp(v)$$

$$\wp'(u+v) = \frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \wp(u+v) + \frac{\wp(u)\wp'(v) - \wp(v)\wp'(u)}{\wp(u) - \wp(v)}.$$

The consequence is the following. If L is a lattice, then we get an elliptic curve $E : y^2 = 4x^3 - g_2(L)x - g_3(L)$.

Now, we have a map $\phi : \mathbb{C} \rightarrow E$ given by

$$\phi(z) = (\wp_L(z), \wp'_L(z)).$$

This map is a homomorphism! If $z, w \in \mathbb{C}$, and $\phi(z) = P$ and $\phi(w) = Q$, then

$$\phi(z+w) = P + Q = \phi(z) + \phi(w).$$

When we write $z + w$ we mean the sum of the two complex numbers z and w , and when we write $P + Q$, we mean addition on the elliptic curve E .

Since $\wp_L(z)$ has a double pole at points in L and $\wp'_L(z)$ has a triple pole at points in L , if $\omega \in L$, then $(\wp_L(z) : \wp'_L(z) : 1) = ((z - \omega)^3 \wp_L(z) : (z - \omega)^3 \wp'_L(z) : (z - \omega)^3)$. Taking a limit as $z \rightarrow \omega$ we get $(0 : a : 0)$, where $a \neq 0$. This makes it sensible to define $\phi(z) = (0 : 1 : 0)$ for $z \in L$.

Theorem. *Assume the notation above. Then,*

- (1) *The map $\phi : \mathbb{C} \rightarrow E(\mathbb{C})$ is a surjective homomorphism.*
- (2) *We have that $\phi(z) = (0 : 1 : 0)$ if and only if $z \in L$.*
- (3) *If g_2 and g_3 are constants so that $4x^3 - g_2x - g_3$ has distinct roots, then there is a lattice L so that $g_2(L) = g_2$ and $g_3(L) = g_3$.*

Interpretation

The theorem above means that we can identify the points on E over $\mathbb{C}$ with complex numbers “mod L ”. That is, we say that $z_1 \equiv z_2 \pmod{L}$ if $z_1 - z_2 \in L$. Then, for each point $P \in E(\mathbb{C})$, we have that $P = \phi(z)$, where z is a unique “residue class” mod L .

Cosets, quotient groups, and the first isomorphism theorem

We need a bit more group theory to explain what’s really going on.

If G is an abelian group and H is a subgroup of G , a coset of H in G is a set of the form

$$a + H = \{a + h : h \in H\},$$

where $a \in G$. Note that we can have $a + H = b + H$ even if $a \neq b$. (This occurs if and only if $a - b \in H$.)

Also, every element of G is contained in some coset, because if $g \in G$, then $g \in g + H = \{g + h : h \in H\}$ because $0 \in H$.

Theorem. *Let G be an abelian group and H a subgroup of G . Then*

- (1) *For any $g \in G$, there is a coset $a + H$ so that $g \in a + H$.*
- (2) *If $a + H$ and $b + H$ are two cosets, then $a + H = b + H$ or $(a + H) \cap (b + H) = \emptyset$.*
- (3) *We can define the operation $(a + H) + (b + H) = (a + b) + H$. This is well-defined binary operation on the set of cosets.*

The last point means that even though a coset can have “more than one name”, the name we use doesn’t matter. More precisely, if $a + H = c + H$ and $b + H = d + H$, then $(a + H) + (b + H) = (c + H) + (d + H)$.

Crazy fact: Let G/H be the set of cosets of H in G . The binary operation defined above makes G/H into a group! The identity of G/H is the coset $0 + H$. The inverse of $a + H$ is $(-a) + H$.

It gets better. There is a map $\phi : G \rightarrow G/H$ given by $\phi(a) = a + H$. This map is a homomorphism, it is surjective. Also, we have that $\phi(a) = 0 + H$ in G/H if and only if $a \in H$.

Definition. *The kernel of a homomorphism $\phi : G \rightarrow K$ is the set of elements in G that map to the identity in K . We denote this set by $\ker \phi$.*

Note: A homomorphism $\phi : G \rightarrow K$ is one-to-one if and only if $\ker \phi = \{1_G\}$.

It gets even better. This construction produces *all* surjective homomorphisms!

Theorem (First isomorphism theorem). *Suppose that G is any abelian group and $\phi : G \rightarrow K$ is a surjective homomorphism. Let $H = \ker \phi$. Then $G/H \cong K$.*

Comment: Quotient groups are the reason for the notation $\mathbb{Z}/n\mathbb{Z}$ for the cyclic group of order n .

(Note: In the non-abelian case, all of the stuff works provided that H is a *normal* subgroup of G . That is, for all $h \in H$ and all $g \in G$, we need ghg^{-1} to be in H . Every normal subgroup is the kernel of some homomorphism.)

Back to elliptic curves

We have the homomorphism $\phi : \mathbb{C} \rightarrow E(\mathbb{C})$. The kernel of ϕ is L , and so $E(\mathbb{C}) \cong \mathbb{C}/L$.

The elements of $\mathbb{C}/L$ are cosets $z + L$. Another way of thinking about this is that we can think of

$$\mathbb{C}/L = \{a\omega_1 + b\omega_2 : 0 \leq a < 1, 0 \leq b < 1\}.$$

Then for $u, v \in \mathbb{C}/L$, we define $u + v$ to be the result of adding u and v , and then subtracting $0, \omega_1, \omega_2$, or $\omega_1 + \omega_2$ to get back into the set above. The set $\{a\omega_1 + b\omega_2 : 0 \leq a < 1, 0 \leq b < 1\}$ is called the *fundamental parallelogram*.

Draw a picture.

Now back to points of finite order.

Q: If $P_1 = (0 : 1 : 0)$, P_2 , P_3 and P_4 are the points of order 2 in $E(\mathbb{C})$, what do they correspond to under the map ϕ ?

A: Well, $0 \in L$ and so $\phi(0) = (0 : 1 : 0) = P_1$. Since the set L is the identity element in $\mathbb{C}/L$, we have that $z \in \mathbb{C}/L$ has order 2 if and only if $2z \in L$ but $z \notin L$. This means that $2z = a\omega_1 + b\omega_2$ and at least one of a or b is odd, so

$$z = \frac{a}{2}\omega_1 + \frac{b}{2}\omega_2.$$

Now, when we “reduce” mod L , we can assume that $0 \leq a/2 < 1$ and $0 \leq b/2 < 1$. This means that $(a/2, b/2) = (1/2, 0)$, $(0, 1/2)$ or $(1/2, 1/2)$. So the points of order 2 in $\mathbb{C}/L$ are $\frac{\omega_1}{2}$, $\omega_2/2$ and $\frac{\omega_1 + \omega_2}{2}$.

Corollary. *If $m \geq 1$, then there are precisely m^2 points in $\mathbb{C}/L$ with order dividing m . Hence, $E[m](\mathbb{C}) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.*

In fact, if P is any point in $E(\mathbb{C})$, then there are precisely m^2 points Q so that $mQ = P$.

Example 1: Let $E : y^2 = x^3 - 13392x - 1080432$.

Magma can determine the lattice corresponding to E . (The command is `Periods(E)`. It gives you two numbers: ω_1 and ω_2 .)

We find that $\omega_1 \approx 0.211535$ and $\omega_2 \approx 0.105767 + 0.243136i$.

The point $P = (168, 1188)$ is a point of order 5 in $E(\mathbb{Q})$, and in fact $E(\mathbb{Q}) = \langle P \rangle$.

We have that $\phi(z) = P$, where $z \approx -0.08461395$.

(The Magma command is `EllipticLogarithm(E! [168, 1188])`. This always returns a value of $z = x + iy$ where $-\omega_1/2 \leq x < \omega_1/2$ and $-\text{Im}(\omega_2)/2 \leq y < \text{Im}(\omega_2)/2$.)

We find that $z \equiv \frac{3}{5}\omega_1$. Now, $2z = -0.1692279 \equiv \frac{6}{5}\omega_1 \equiv \frac{1}{5}\omega_1 \pmod{L}$.

It turns out that $\phi(\omega_2/5)$ maps to the point

$$(-247.623 - 255.831i, 6338.187 - 2121.300i).$$

(You can get this with `EllipticExponential(E, Periods(E) [2]/5)`.)

With more decimals, we can actually see that this is the point

$$(72\zeta^3 - 72\zeta^2 - 180\zeta - 192, -1404\zeta^3 - 2160\zeta^2 - 648\zeta + 486),$$

where $\zeta = e^{2\pi i/5}$. The x and y coordinates of this point satisfy a degree 4 polynomial with integer coefficients. These polynomials can be found by using the group law on the elliptic curve.

Example 2: Let $E : y^2 = x^3 - 16x + 16$.

We have $\omega_1 \approx 1.4967293231159798149$ and $\omega_2 \approx 1.22569i$. The point $P = (4 : -4 : 1)$ is a point of infinite order on E .

We have $\phi(z) = P$, where $z \approx 0.56713660783$.

Claim: z/ω_1 is irrational.

If n is a positive integer, and nz/ω_1 is close to an integer k , then nz is close to $k\omega_1$. Since $\wp_L(z)$ has a pole at points in L , this means that $\wp_L(nz)$ will be very large.

For example, we find that

$$351z = 132.9999661\omega_1,$$

and this means that $351P \approx (390170700.0433 : 7706940023561.8368 : 1)$.